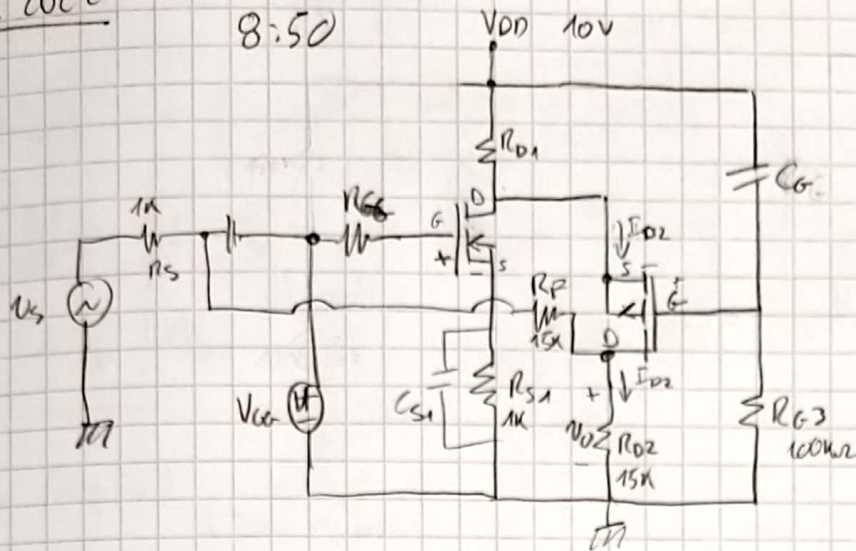


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ES 1



1) Value of R_{D1} such that $V_o = 2V$ is as:

From on R_{D1} $V_{GS} = \frac{V_{DD} \cdot R_{G2}}{R_{G2} + R_{G1}} = 4V$ $R_{G2} = R_{D1} / 11 R_{G2} = 60k\Omega$

$$I_{D1} = I_{D1} + I_{D2} \quad \boxed{I_{D2} = \frac{V_o}{R_{D2} \parallel (R_L + R_S)} = 0.258mA}$$

$$V_{G2} = 0V$$

$$I_{D2} = I_{DSS2} \left(1 - \frac{V_{GS}}{V_{T2}} \right)^2 = I_{DSS2} \left(1 + \frac{V_{GS}^2}{V_{T2}^2} - \frac{2V_{GS}}{V_{T2}} \right)$$

$$\frac{I_{D2}}{I_{DSS2}} = 1 + \frac{V_{GS}^2}{V_{T2}^2} - \frac{2V_{GS}}{V_{T2}}$$

$$V_{GS}^2 \left(\frac{1}{V_{T2}^2} \right) + V_{GS} \left(-\frac{2}{V_{T2}} \right) + 1 - \frac{I_{D2}}{I_{DSS2}} = 0$$

$$V_{GS2} = -2.5V$$

$$\boxed{V_{GS2} = -3.8V} \quad V_{S2} = -V_{GS2} = 3.8V$$

$$V_{DS2} = V_o - V_{S2} = 2 - 3.8 = -1.8V$$

$$V_{S2} = -3.57$$

$$\frac{V_{DD} - V_{S2}}{R_{D1}} = I_{D1} + I_{D2}$$

$$R_{D1} = I_{DSS1} \left(1 - \frac{V_{GS1}}{V_{T1}} \right)^2 \quad V_{GS1} = V_{GG} - R_{S1} I_{D1}$$

$$I_{D1} = \frac{V_{GS} - V_{GS1}}{R_{S1}}$$

$$\frac{V_{GG} - V_{GS}}{I_{DSS1} R_{S1}} = 1 + \frac{V_{GS}^2}{V_{T1}^2} - \frac{2V_{GS}}{V_{T1}}$$

$$\frac{V_{GS}}{3} - \frac{V_{GS1}}{2k\Omega} = 1 + \frac{V_{GS}^2}{4} - V_{GS1}$$

$$V_{GS1} \left(\frac{1}{k\Omega} \right) + V_{GS1} \left(-1 + \frac{1}{2k\Omega} \right) + 1 - \frac{V_{GS}}{3} = 0$$

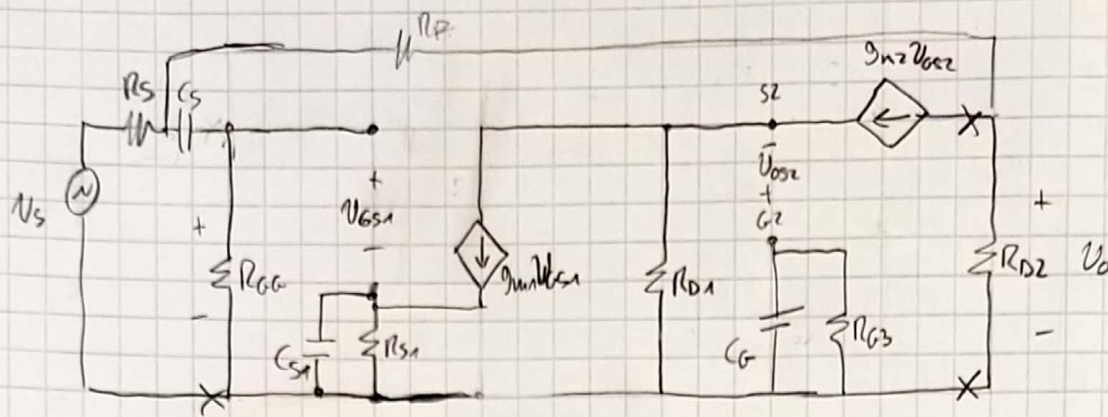
$$\boxed{V_{GS1} = 3.17V}$$

$$\boxed{I_{D1} = 0.82mA}$$

$$R_{D1} = \frac{V_{DD} - V_{S2}}{I_{D1} + I_{D2}} = 5.75k\Omega$$

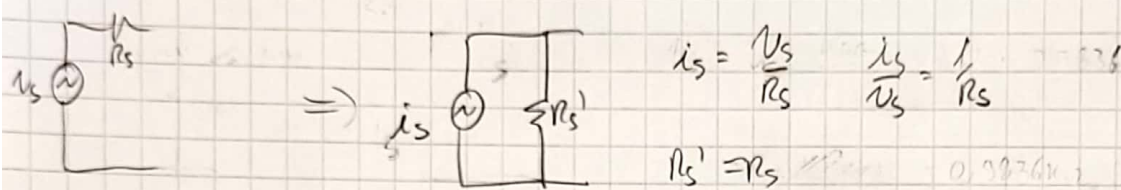
$$V_{DS1} = V_{S2} - V_{GS1} = 2.97V$$

3) Voltage gain $A_v = v_o/v_s$

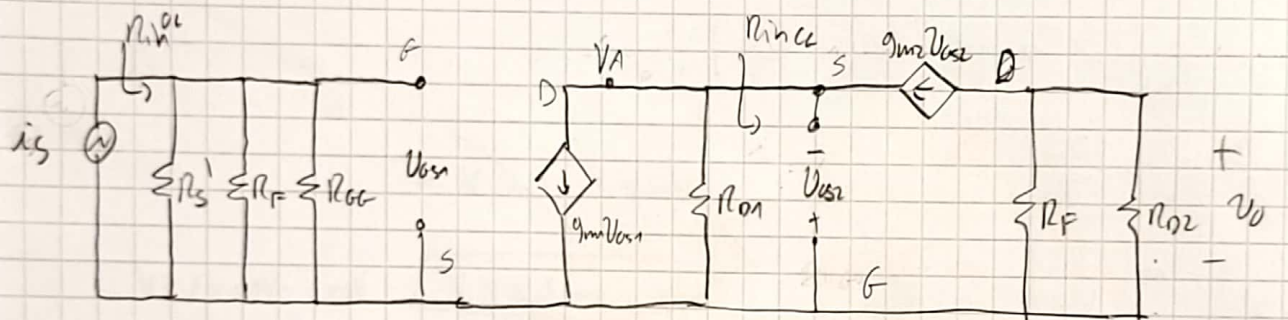


mixing: $v_{gs1} = R_{G1} (i_s + i_F) \Rightarrow$ current mixing (applies Norton as input)

sampling: short $v_o \rightarrow v_o$ feedback $\rightarrow$ voltage sampling



loading effects: $R_{in} = R_{o1} = R_F$ $\beta = -1/R_F$ $g_{m1} = \frac{2I_{D1}}{V_{GS1} - V_{th1}} = 1.62 \text{ mS}$ $g_{m2} = 0.665 \text{ mS}^{-1}$



$$A_{vol} = \frac{v_o}{i_s} = A_{VGS} \cdot A_{VCS} = \frac{v_o}{v_A} \cdot \frac{v_A}{v_{gs1}} \cdot \frac{v_{gs1}}{i_s}$$

$$R_{inCG} = \frac{v_m}{i_m} = \frac{-v_{gs2}}{-g_{m2}v_{gs2}} = \frac{1}{g_{m2}}$$

$$A_{VGS} = g_{m2} (R_F // R_{o2}) = 4.8375$$

$$A_{VCS} = -g_{m1} (R_{o1} // R_{inCG}) = -1.73$$

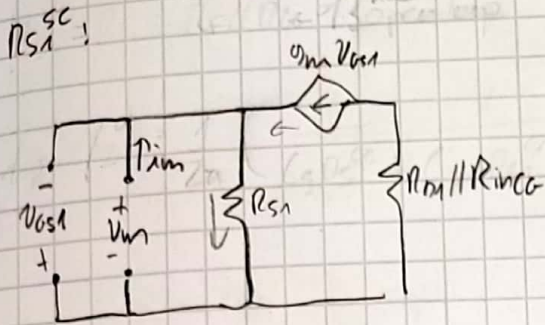
$$\frac{v_{gs1}}{i_s} = R_{in}^a = R_s' // R_F // R_{G1} = 0.923 \text{ k}\Omega$$

$$A_{vol} = -7.72 \text{ k}\Omega \quad A_v^F = \frac{A_v^a}{1 + \beta A_{vol}} = \boxed{-5.1 \text{ k}\Omega} \quad A_v = A_v^F \cdot \frac{1}{R_s} = \boxed{-5.01 \text{ V/V}}$$

$$1) R_{in}^{OL} = 0.923 \text{ k}\Omega \quad R_{in}^F = R_{in}^a / (1 + \beta A_{vol}) = 0.61 \text{ k}\Omega$$

$$R_{in} = \frac{1}{\frac{1}{R_{in}^F} - \frac{1}{R_s}} = \boxed{1.57 \text{ k}\Omega}$$

5) Low freq limit using SETC. all 3 caps are equal
 feedback reduces R_{in} and so reduces
 $R_S^{sc} \rightarrow \frac{1}{C_S R_S^{sc}}$ increase $\rightarrow$ not on to apply feedback
 but anyway...



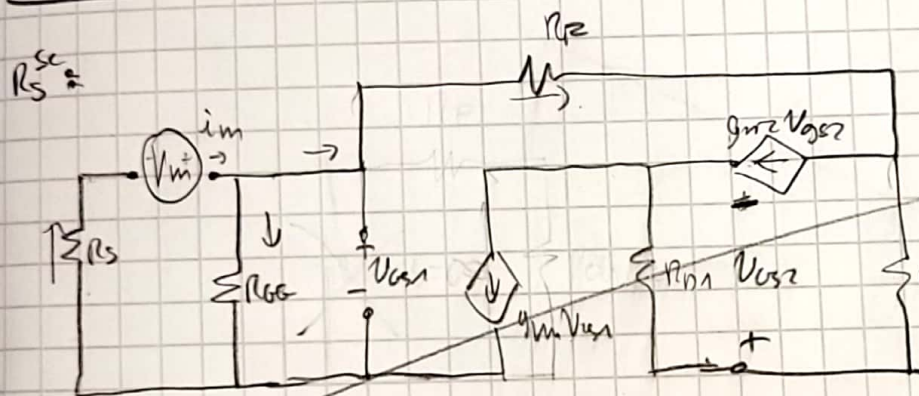
$$i_{in} = \frac{V_m}{R_{S1}} - g_m V_{GS1}$$

$$V_m = -V_{GS1}$$

$$\rightarrow i_{in} = V_m \left(\frac{1}{R_{S1}} + g_m \right)$$

$$\frac{V_m}{i_{in}} = R_{S1}^{sc} = \frac{1}{\frac{1}{R_{S1}} + g_m} = 614 \Omega$$

$$R_G^{sc} = R_{GS} = 100 k\Omega$$



$$i_{in} = i_{G_S} + i_F$$

$$i_F = g_m V_{GS2} + V_o$$

$$V_m = V_o + R_F i_F + R_S i_{in}$$

$$i_{GG} = \frac{V_m}{R_S + R_{GG}}$$

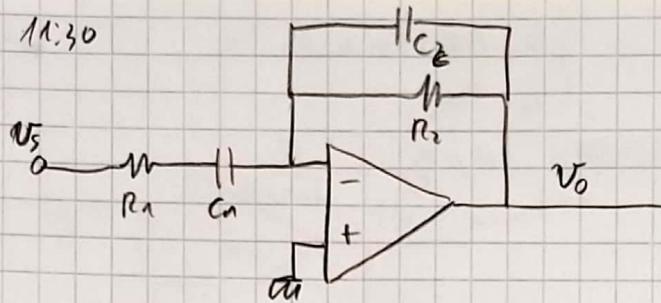
$$i_{in} = \frac{V_m}{R_S + R_{GG}} + g_m V_{GS2} + V_o = \frac{V_m}{R_S + R_{GG}} + g_m$$

$$R_S^{sc} = R_S + R_F + R_{GG} = 60,93 k\Omega$$

$$f_L^{oc} = 38,86 \text{ Hz}$$

$$f_L^F = 23,65 \text{ Hz}$$

Ex 2 11:30



$$A(s) = \frac{A_{d0}}{\left(1 + s/\omega_0\right) \left(1 + s/\omega_1\right)}$$

$\omega_0 = 10 \text{ rad/s}$

$\omega_1 = 500 \text{ rad/s}$

1) $C_2 = \text{open}$

Ideal voltage gain:

$$A_v(s) = -\frac{Z_2(s)}{Z_1(s)} = -\frac{R_2}{R_1 + \frac{1}{sC_1}} = \frac{-R_2}{\frac{1 + sC_1 R_1}{sC_1}} = \frac{-sC_1 R_2}{1 + sC_1 R_1}$$

2) estimation of circuit PM:

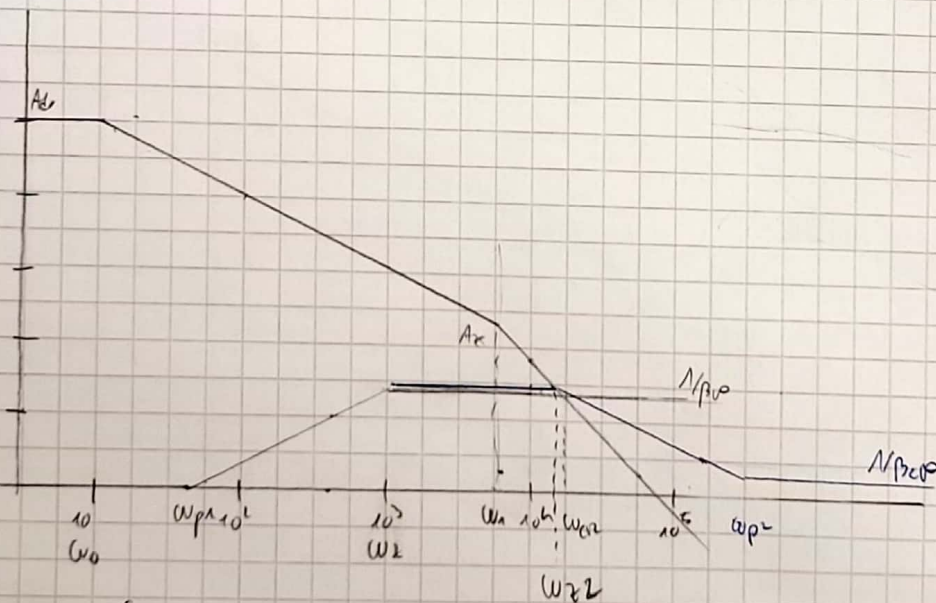
$$T = \frac{V_r}{V_e} = A_d \frac{Z_1}{Z_2 + Z_1} = A_d(s) \cdot \frac{\frac{1 + sC_1 R_1}{sC_1}}{\frac{R_2 + 1 + \frac{sC_1 R_1}{sC_1}}{sC_1}} = A_d(s) \cdot \frac{1 + sC_1 R_1}{1 + sC_1 (R_2 + R_1)}$$

$$\beta_0 = 1 = 0 \text{ dB} = 1/\beta_0$$

$$\beta_{\infty} = \frac{R_1}{R_2 + R_1} = 1/23$$

$$\omega_p = \frac{1}{C_1(R_1 + R_2)} = 43.5 \text{ rad/s} \quad \omega_z = \frac{1}{C_1 R_1} = 10^3 \text{ rad/s}$$

$$1/\beta_{\infty} = 23 = 27.23 \text{ dB}$$



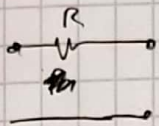
$$\begin{cases} A_{d0} \omega_0 = A_x \omega_1 \\ A_x \omega_1^2 = \frac{1}{\beta_{\infty}} \omega_{cr}^2 \end{cases} \quad \begin{cases} A_x = A_{d0} \omega_0 / \omega_1 \\ A_{d0} \omega_0 \omega_1 = \omega_{cr}^2 / \beta_{\infty} \end{cases}$$

$$\omega_{cr} = \sqrt{A_{d0} \omega_0 \omega_1 \beta_{\infty}} = 1.475 \cdot 10^4 \text{ rad/s}$$

$$\varphi_m = 180 - \arctan\left(\frac{\omega_{cr}}{\omega_0}\right) - \arctan\left(\frac{\omega_{cr}}{\omega_1}\right) - \arctan\left(\frac{\omega_{cr}}{\omega_p}\right) + \arctan\left(\frac{\omega_{cr}}{\omega_z}\right) = 15.05^\circ$$

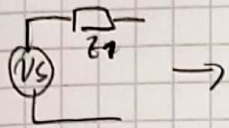
3) block diagram from feedback theory

current mixing voltage sampling



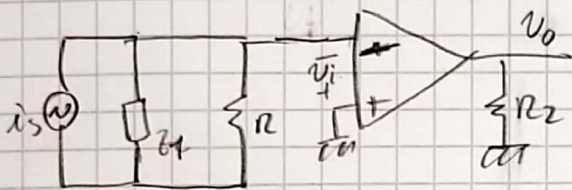
$$\beta = -\frac{1}{R} \quad R_{if} = R_{oi} = R$$

motion on input



$$i_s = V_s / Z_1$$

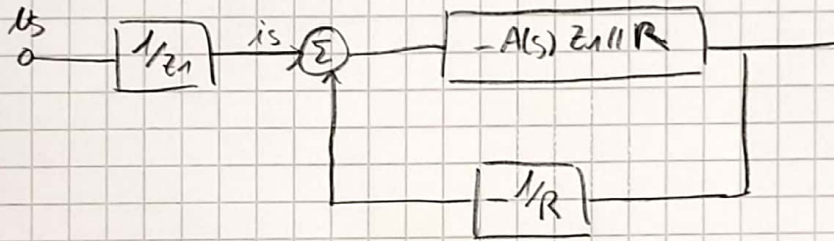
$$Z_2 = R_2$$



$$A_{OL} = \frac{V_o}{i_s} = \frac{V_o}{V_i} \cdot \frac{V_i}{i_s} = -A(s) \cdot Z_1 // R$$

$$Z_1 = R_1 + \frac{1}{sC_1} = \frac{1 + sC_1 R_1}{sC_1}$$

$$K_S = \frac{i_s}{V_s} = \frac{1}{Z_1}$$



$$K_S = \frac{sC_1}{1 + sC_1 R_1}$$

5) max cc for same freq resp up to ω_{cr}

$$\text{now } Z_2 = \frac{R_2}{1 + sC_2 R_2}$$

$$\beta = \frac{Z_1}{Z_2 + Z_1} = \frac{\frac{1 + sC_1 R_1}{sC_1}}{\frac{R_2}{1 + sC_2 R_2} + \frac{1 + sC_1 R_1}{sC_1}} = \frac{1 + sC_1 R_1}{sC_1} \cdot \frac{1 + sC_2 R_2}{sC_1 R_2 + (1 + sC_1 R_1)(1 + sC_2 R_2)}$$

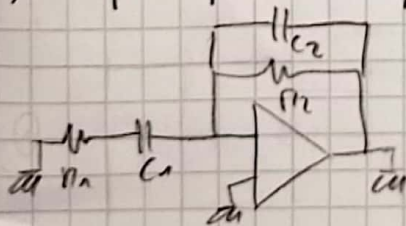
$$= \frac{(1 + sC_1 R_1)(1 + sC_2 R_2)}{1 + sC_2 R_2 + sC_1 R_1 + sC_1 R_2 + s^2 C_1 C_2 R_1 R_2}$$

no need for calculations

β introduces $\rightarrow$ zero and $\rightarrow$ pole

the zero is found at $\omega_z = \frac{1}{C_2 R_2}$ and we place it at ω_{cr} $C_2 = \frac{1}{R_2 \omega_{cr}} = \boxed{308 \text{ pF}}$

6) new pm: β has a LF pole and a HF pole, the latter falling well above ω_{cr}



C_1 introduces the LF pole that i have $\omega_{p1} = 63.5 \text{ rad/s}$

C_2 introduces a pole whose contribution to ϕ_m is negligible, but anyhow

$$\omega_{z2} = \omega_{cr} \text{ for design} \rightarrow \frac{1}{\beta_0} \omega_{cr} = 1 \omega_{pe} \rightarrow \omega_{pe} = 23 \omega_{z2} = 3.6 \cdot 10^5 \text{ rad/s}$$

$$(\beta_{cr} = 1)$$

$$\beta_c(0) = 1 \quad \beta_{cr} = 1$$

$$\phi_m = 15.05 - \arctan\left(\frac{\omega_{cr}}{\omega_{p2}}\right) + \arctan\left(\frac{\omega_{cr}}{\omega_{z2}}\right) = \boxed{57.6^\circ}$$

$\rightarrow$ negligible, ω_{cr}